



# **NATIONAL OCCUPATIONAL STANDARD FOR AQUACULTURAL ECONOMIST**

**NOS.AE.01  
FIRST EDITION**

## APPROVING AUTHORITY

This National Occupational Standard has been prepared and published under the authority of the Zambia Qualifications Authority Board on 19<sup>th</sup> May 2022.

## ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through various sector specific National Occupational Standards Development Teams (NOSDTs).

## REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards shall be revised after every **5 years**, or whenever necessary, by the issue of either amendments or of revised editions. It is important that users of National Occupational Standards (NOS) ascertain that they are in possession of the latest amendments or editions.

## NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Agriculture National Occupational Standards Development Team, upon which the following organisations were represented:

1. Agricultural Institution of Zambia (AIZ)
2. Aquaculture Development Association of Zambia
3. CropLife Zambia
4. Ministry of Fisheries and Livestock (Department of Fisheries)
5. Golden Valley Agricultural Research Trust (GART)
6. Katete College of Agricultural Marketing
7. Ministry of Agriculture (Department of Agriculture)
8. Mulungushi University (MU)
9. Natural Resources Development College (NRDC)
10. University of Zambia (UNZA)
11. Veterinary Council of Zambia
12. Zambia Agriculture Research Institute (ZARI)
13. Zambia National Farmers Union (ZNFU)
14. Zambia Seed Trade Association (ZASTA)
15. Zambia Qualifications Authority (ZAQA)

## ACKNOWLEDGEMENT

The Zambia Qualifications Authority would like to acknowledge the invaluable support of the following stakeholders that participated in the development of this National Occupational Standard:

1. Dr. Amy Kingdom (Zambia National Farmers Union - ZNFU)
2. Dr. Batiseba Tembo (Zambia Agriculture Research Institute - ZARI)
3. Dr. Elias Kantashula (University of Zambia - UNZA)
4. Dr. Joself M. Chileshe (University of Zambia - UNZA)
5. Maj. Nelson kaluba (Rtd) (Aquaculture Development Association of Zambia)
6. Mr. Alex Dennis Chilala (Zambia Agriculture Research Institute – ZARI)
7. Mr. Davis Mwanza (Natural Resources Development College - NRDC)
8. Mr. Enerst Muzukutwa (CropLife Zambia/ZASTA)
9. Mr. Johnstone Mfula (Ministry of Fisheries and Livestock (Department of Fisheries)
10. Mr. Malumo Nawa (Ministry of Agriculture (Department of Agriculture)
11. Mr. Masautso E. Sakala (Natural Resources Development College - NRDC)
12. Mr. Mathews Ngosa (Agricultural Institution of Zambia (AIZ))
13. Mr. Shadreck C. Mubanga (Golden Valley Agricultural Research Trust - GART)
14. Mr. Stanford Phiri (Katete College of Agricultural Marketing)
15. Mr. Joseph T. Mwale (Mulungushi University)
16. Mrs. Precious Hamusute (Veterinary Council of Zambia)
17. Ms. Eva N. Kaonga (University of Zambia - UNZA)
18. Ms. Sylvia H. N'gandu (University of Zambia - UNZA)
19. Mr. Fidelis Cheelo (Zambia Qualifications Authority)
20. Miss Womba Soneka (Zambia Qualifications Authority)
21. Mr. Modest Hamalabbi (Zambia Qualifications Authority)

## TABLE OF CONTENTS

|                                                                      |    |
|----------------------------------------------------------------------|----|
| FOREWORD .....                                                       | iv |
| JUSTIFICATION .....                                                  | iv |
| ACRONYMS AND ABBREVIATIONS .....                                     | v  |
| GLOSSARY OF TERMS .....                                              | vi |
| 1. OVERVIEW .....                                                    | 1  |
| 2. SCOPE .....                                                       | 2  |
| 3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES) .....          | 2  |
| 4. UNITS AND ELEMENTS .....                                          | 2  |
| 5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS .....                   | 18 |
| 6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER .....       | 18 |
| 7. WORKING CONDITIONS/ENVIRONMENT .....                              | 18 |
| 8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE ..... | 18 |
| 9. PHYSICAL DEMANDS ON THE BODY .....                                | 19 |
| ANNEX A .....                                                        | 20 |
| ANNEX B .....                                                        | 21 |

## FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to ***“provide for the development and implementation of a national qualifications framework; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***.

Among other functions, ZAQA is responsible for ***“determining national standards for any occupation”***, through various sector specific National Occupational Standards Development Teams (NOSDTs) of experts composed of representation from appropriate authorities, government departments, industry, academia, regulators, consumer associations and non-governmental organizations, etc.

This National Occupational Standard (NOS) has been developed by the Agriculture National Occupational Standards Development Team in accordance with the procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula and learning programmes, in various sectors where the occupation exists. In the Agriculture sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

## JUSTIFICATION

Aquaculture is a fast growing industry in Zambia owing to the fact that there is a deficit of about 70,000 metric tonnes forcing the country to import fish from countries like China. Studies in different countries have shown the great importance of integrated fish culture in augmenting capture fisheries production, generating employment, producing adequate food for the population, and generally improving the economy of the rural areas.

To adequately maximize benefits from this growing industry, there is need to understand the dynamics and profitability of aquaculture, hence the need for an Aquacultural Economist to inform policy and decision making in the industry. The importance of economic analysis is emphasized since it provides a basis not only in the decision making of the individual farmer, but also in the formulation of aquaculture policies. Thus, greater attention should be focused on the improvement of economic data for analysis.

## ACRONYMSAND ABBREVIATIONS

|       |                                                  |
|-------|--------------------------------------------------|
| AE    | Aquacultural Economist                           |
| CS    | Core Skill                                       |
| NOS   | National Occupational Standard                   |
| NOSDT | National Occupational Standards Development Team |
| OK    | Organizational Knowledge                         |
| PC    | Performance Criteria                             |
| PS    | Professional Skill                               |
| RK    | Regulatory Knowledge                             |
| RPL   | Recognition of Prior Learning                    |
| TK    | Technical Knowledge                              |
| ZAQA  | Zambia Qualifications Authority                  |
| ZQF   | Zambia Qualifications Framework                  |

## GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

**Core Skills/Generic Skills:** are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

**Function:** is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.

**Job Title:** defines a unique set of functions that together form a unique employment opportunity in an organisation.

**Knowledge and Understanding:** are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

**National Occupational Standards (NOS):** are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

**National Occupational Standards (NOS) Code:** is a unique reference code that identifies a NOS.

**National Occupational Standards Development Team (NOSDT):** means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

**Occupation:** is a set of job roles, which perform similar/related set of functions in an industry.

**Organisational Context:** includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

**Performance Criteria:** are statements that together specify the standard of performance required when carrying out a task.

**Scope:** is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

**Sector:** is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.

**Sub Sector:** is derived from a further breakdown based on the characteristics and interests of its components.

**Technical Knowledge:** is the specific knowledge needed to accomplish specific designated responsibilities.

**Unit Title:** gives a clear overall statement about what the incumbent should be able to do.

## 1. OVERVIEW

This is an introductory section providing a brief summary and specific information or commentary about the content of the NOS and the targeted sector and occupation to help the user judge whether it is relevant to them.

|                                                                              |                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>                                                              | NOS.AE.01                                                                                                                                                                                                                                                                      |
| <b>Occupation</b>                                                            | Aquacultural Economist                                                                                                                                                                                                                                                         |
| <b>Job Title</b>                                                             | Aquacultural Economist                                                                                                                                                                                                                                                         |
| <b>Job Description</b>                                                       | An Aquacultural Economist is responsible for the development & implementation of programmes for the aquaculture industry. Further, the job holder is expected to do Economic analysis & modelling of aquaculture and Capacity development in support for aquaculture industry. |
| <b>Job Purpose</b>                                                           | An Aquacultural Economist is responsible for providing economic advice to government, assessments and analysis of aquaculture data, cost-benefit analysis and business skills to stakeholders and the private sector engaged in aquacultural activities.                       |
| <b>ZQF Level</b>                                                             | 7                                                                                                                                                                                                                                                                              |
| <b>Sector</b>                                                                | Agriculture                                                                                                                                                                                                                                                                    |
| <b>Sub sector</b>                                                            | Aquaculture Economics                                                                                                                                                                                                                                                          |
| <b>Other Economic Sector(s) in which the Occupation is Practiced</b>         | Financial and insurance sector;<br>Commercial and manufacturing sector;<br>Various governmental ministries; and<br>self-employment                                                                                                                                             |
| <b>Other Similar Jobs that can be performed by an Aquacultural Economist</b> | Capture fisheries<br>Business Management<br>Monitoring and evaluation;<br>Project managers;<br>Self-employment                                                                                                                                                                 |
| <b>Minimum Educational Job Entry Qualification(s)</b>                        | Degree in Aquaculture and Fisheries Sciences or equivalent expertise                                                                                                                                                                                                           |
| <b>Practicing License Requirements (if any)</b>                              | N/A                                                                                                                                                                                                                                                                            |
| <b>Training/RPL</b>                                                          | Fisheries and Aquaculture Sciences<br>Aquaculture Economics                                                                                                                                                                                                                    |
| <b>Minimum Job Entry Age</b>                                                 | 21 years                                                                                                                                                                                                                                                                       |
| <b>Prior Experience</b>                                                      | Demonstrated professional experience in the aquaculture value chain of at least 2 years                                                                                                                                                                                        |
| <b>Performance Criteria</b>                                                  | As described in the Units under Section 4                                                                                                                                                                                                                                      |

## **2. SCOPE**

This National Occupational Standard specifies the fundamental knowledge and understanding, skills and competences that an Aquacultural Economist must possess to be successful in his/her job role. The job scope is applicable to Aquacultural Economists working in public or private organisations or self-employed.

## **3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)**

This job requires an individual to possess:

- Creativity
- Problem solving skills
- Analytical skills
- Mathematics skills
- Integrity and respect for confidentiality
- Interpersonal skills
- Commercial Awareness
- Attention to details
- Ability to communicate effectively and clearly
- Self-motivated and team worker
- Ability to plan and prioritize,
- Quality consciousness

## **4. UNITS AND ELEMENTS**

This National Occupational Standard is divided into 7 units representing the tasks that a job holder should undertake in his/her day to day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

**UNIT 1** [This unit is about basic principles in economics and aquaculture production].

|                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit No.</b>                                                                             | <b>01</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Unit Title</b>                                                                           | <b>Basic principles in economics and aquaculture production</b>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Description</b>                                                                          | This unit is about understanding the fundamental principles of economics and aquaculture production.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Scope</b>                                                                                | <p>This unit covers the following:</p> <ul style="list-style-type: none"> <li>• Fundamental micro and macro-economic principles in aquaculture</li> <li>• Fundamentals of Aquaculture and Fisheries Science</li> <li>• Concepts of Sustainable Aquaculture Development</li> <li>• Assessment of costs and social benefits in aquaculture</li> </ul>                                                                                                                                 |
| <b>Performance Criteria (PC) w.r.t. the Scope</b>                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Element</b>                                                                              | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Fundamental micro and macro-economic principles in aquaculture</b>                       | <p>To be competent, the individual must be able to:</p> <p>PC1. Apply micro-economics concepts</p> <p>PC2. Practice fish farming and its production processes (costs, revenues and profit level).</p> <p>PC3. Manage the operation at the farm</p>                                                                                                                                                                                                                                  |
| <b>Fundamentals of Aquaculture and Fisheries Science</b>                                    | <p>To be competent, the individual must be able to:</p> <p>PC4. Synthesize the information carried in aquaculture and fisheries science</p> <p>PC5. Differentiate between aquaculture and fisheries production.</p> <p>PC6. Apply concepts to practical situations in Aquaculture and related fields</p>                                                                                                                                                                            |
| <b>Concepts of Sustainable Aquaculture Development</b>                                      | <p>To be competent, the individual must be able to:</p> <p>PC7. Apply concepts of sustainable aquaculture and the cultivation of aquatic organism for commercial purposes that generate profit.</p> <p>PC8. Interpret the importance of aquaculture in the fisheries industry.</p> <p>PC9. Demonstrate the importance of Aquaculture and its contribution to GDP</p>                                                                                                                |
| <b>Assessment of costs and social benefits in aquaculture</b>                               | <p>To be competent, the individual must be able to undertake:</p> <p>PC10: Assessment of social benefits and costs of an aquaculture project.</p>                                                                                                                                                                                                                                                                                                                                   |
| <b>Knowledge and Understanding (K)</b>                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>A. Organisational Context (Knowledge of the company/ organisation and its processes)</b> | <p>To be competent, the individual must be able to:</p> <p>Ok1: Define the basic concepts of micro-economics</p> <p>Ok2: Describe fish farming and its production processes (costs, revenues and profit level).</p> <p>Ok3: Describe management measures and elements of operation at the farm level</p> <p>Ok4: Explain the four basic market models</p> <p>Ok5: Measure inflation and employment rates</p> <p>Ok6: Measure gross national product using the spending approach</p> |

|                                                                   |                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                   | <p>OK7: Describe investment interest rate and exchange rate</p> <p>OK8: Describe fiscal policy, monetary policy and international trade.</p>                                                                                                                                        |
| <b>B. Technical Knowledge</b>                                     | <p>The individual on the job must demonstrate knowledge and understanding of:</p> <p>TK1. Basic knowledge on aquacultural production and economies</p> <p>TK2. Basic knowledge on economic models</p>                                                                               |
| <b>C. Regulatory context (Knowledge of Rules and Regulations)</b> | <p>The individual on the job must demonstrate knowledge and understanding of :</p> <p>RK1. Government regulatory requirements in the aquacultural sector</p>                                                                                                                        |
| <b>Skills (S)</b>                                                 |                                                                                                                                                                                                                                                                                     |
| <b>A. Core Skills/ Generic Skills</b>                             | <b>Writing Skills</b>                                                                                                                                                                                                                                                               |
|                                                                   | <p>The individual on the job must be able to:</p> <p>CS1. Write in English and give simple concise instructions.</p>                                                                                                                                                                |
|                                                                   | <b>Reading Skills</b>                                                                                                                                                                                                                                                               |
|                                                                   | <p>The individual on the job must be able to:</p> <p>CS2. Read and interpret internal/external documents.</p> <p>CS3. Read and understand manuals, memos, other company documents.</p> <p>CS4. Read from different sources- books, screens in machines and signage.</p>             |
|                                                                   | <b>Oral Communication (Listening and Speaking skills)</b>                                                                                                                                                                                                                           |
|                                                                   | <p>The individual on the job must be able to :</p> <p>CS5. Express statements or information clearly so that others can hear and understand.</p> <p>CS6. Participate in and understand the main points of simple discussions.</p> <p>CS7. Respond appropriately to any queries.</p> |
| <b>B. Professional Skills</b>                                     | <b>Decision Making</b>                                                                                                                                                                                                                                                              |
|                                                                   | <p>The individual on the job must be able to:</p> <p>PS1. Follow organization rule-based decision-making process.</p> <p>PS2. Take decision with systematic course of actions and/or response.</p>                                                                                  |
|                                                                   | <b>Plan and Organise</b>                                                                                                                                                                                                                                                            |
|                                                                   | <p>The individual on the job must be able to:</p> <p>PS3. Plan and organise work to meet deadlines.</p> <p>PS4. Work constructively and collaboratively with others.</p>                                                                                                            |
|                                                                   | <b>Customer Centricity</b>                                                                                                                                                                                                                                                          |
|                                                                   | <p>The individual on the job must be able to:</p> <p>PS5. Follow the code of conduct.</p> <p>PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.</p>                                                                          |

|  |                                                                                                                                                                                                                                                                          |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Problem Solving and Decision Making</b>                                                                                                                                                                                                                               |
|  | The individual on the job must be able to:<br>PS7 Recognize problems and search for solutions.<br>PS8. Choose best methods to complete assigned tasks.<br>PS9. Approach relevant authority when required.<br>PS10. Judiciously use common sense in day to day activities |
|  | <b>Analytical Thinking</b>                                                                                                                                                                                                                                               |
|  | The individual on the job must be able to:<br>PS11. Apply domain knowledge, observations and data to select course of action to perform tasks                                                                                                                            |
|  | <b>Critical Thinking</b>                                                                                                                                                                                                                                                 |
|  | The individual on the job must be able to:<br>PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities.<br>PS13. Ask relevant questions for better understanding.                                       |

**UNIT 2 [This unit is about the aquaculture production economy and integrated fish farming].**

|                                                                      |                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit No.</b>                                                      | <b>02</b>                                                                                                                                                                                                                                                                     |
| <b>Unit Title</b>                                                    | <b>Aquaculture Production Economics</b>                                                                                                                                                                                                                                       |
| <b>Description</b>                                                   | This unit is about analysing the relationship between the inputs, and production as well as human resource or labour.                                                                                                                                                         |
| <b>Scope</b>                                                         | This unit covers the following: <ul style="list-style-type: none"> <li>• Aquaculture production and concept of Integrated fish farming</li> <li>• Production Economics</li> <li>• Aquacultural extension</li> </ul>                                                           |
| <b>Performance Criteria (PC) w.r.t. the Scope</b>                    |                                                                                                                                                                                                                                                                               |
| <b>Element</b>                                                       | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                              |
| <b>Aquaculture production and concept of Integrated fish farming</b> | To be competent, the individual must be able to:<br>PC1. Explain production requirements of aquaculture<br>PC2. Solve the challenges of aquaculture production.<br>PC3. Apply the concept of integrated fish farming<br>PC4. Explain the limitations of pig and fish farming. |
| <b>Production Economics</b>                                          | To be competent, the individual must be able to:<br>PC5. Apply the key concepts of production economics in an aqua business setup.<br>PC6. Demonstrate the use of mathematical applications in solving production economics problems in an aqua business.                     |
| <b>Aquacultural extension</b>                                        | To be competent, the individual must be able to:<br>PC7. Explore current trends in thinking about and doing aquacultural extension.                                                                                                                                           |

| <b>Knowledge and Understanding (K)</b>                                                     |                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A. Organisational Context (Knowledge of the company/organisation and its processes)</b> | The individual on the job must demonstrate knowledge and understanding of:<br>OK1. Company regulations and global best practices<br>OK2. Guidelines for liaising with industry players<br>OK3. Code of business conduct<br>OK4. Job responsibilities/duties                                                                                            |
| <b>B. Technical Knowledge</b>                                                              | The individual on the job must demonstrate knowledge and understanding of:<br>TK1. Basic knowledge on aquacultural production and economies<br>TK2. Basic knowledge on economic models                                                                                                                                                                 |
| <b>C. Regulatory context (Knowledge of Rules and Regulations)</b>                          | The individual on the job must demonstrate knowledge and understanding of :<br>RK1. Government regulatory requirements in the agricultural sector.                                                                                                                                                                                                     |
| <b>Skills (S)</b>                                                                          |                                                                                                                                                                                                                                                                                                                                                        |
| <b>A. Core Skills/ Generic Skills</b>                                                      | <b>Writing Skills</b>                                                                                                                                                                                                                                                                                                                                  |
|                                                                                            | The individual on the job must be able to:<br>CS1. Write in English and give simple concise instructions.                                                                                                                                                                                                                                              |
|                                                                                            | <b>Reading Skills</b>                                                                                                                                                                                                                                                                                                                                  |
|                                                                                            | The individual on the job must be able to:<br>CS2. Read and interpret internal/external documents.<br>CS3. Read and understand manuals, memos, other company documents.<br>CS4. Read from different sources- books, screens in machines and signage.<br>CS5. Understand the various colour codes, nomenclature and acronyms related to the profession. |
|                                                                                            | <b>Oral Communication (Listening and Speaking skills)</b>                                                                                                                                                                                                                                                                                              |
| <b>B. Professional Skills</b>                                                              | The individual on the job must be able to :<br>CS6. Express statements or information clearly so that others can hear and understand.<br>CS7. Participate in and understand the main points of simple discussions.<br>CS8. Respond appropriately to any queries.                                                                                       |
|                                                                                            | <b>Decision Making</b>                                                                                                                                                                                                                                                                                                                                 |
|                                                                                            | The individual on the job must be able to:<br>PS1. Follow organization rule-based decision-making process.<br>PS2. Take decision with systematic course of actions and/or response.                                                                                                                                                                    |
|                                                                                            | <b>Plan and Organise</b>                                                                                                                                                                                                                                                                                                                               |
|                                                                                            | The individual on the job must be able to:<br>PS3. Plan and organise work to meet deadlines.<br>PS4. Work constructively and collaboratively with others.                                                                                                                                                                                              |
| <b>C. Customer Centricity</b>                                                              | <b>Customer Centricity</b>                                                                                                                                                                                                                                                                                                                             |
|                                                                                            | The individual on the job must be able to:<br>PS5. Follow the code of conduct.                                                                                                                                                                                                                                                                         |

|  |                                                                                                                                                                                                                                                                          |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | PS6.Manage relationships with customers with intent on satisfying its requirements for service delivery.                                                                                                                                                                 |
|  | <b>Problem Solving and Decision Making</b>                                                                                                                                                                                                                               |
|  | The individual on the job must be able to:<br>PS7 Recognize problems and search for solutions.<br>PS8. Choose best methods to complete assigned tasks.<br>PS9. Approach relevant authority when required.<br>PS10. Judiciously use common sense in day to day activities |
|  | <b>Analytical Thinking</b>                                                                                                                                                                                                                                               |
|  | The individual on the job must be able to:<br>PS11. Apply domain knowledge, observations and data to select course of action to perform tasks                                                                                                                            |
|  | <b>Critical Thinking</b>                                                                                                                                                                                                                                                 |
|  | The individual on the job must be able to:<br>PS12.Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities.<br>PS13. Ask relevant questions for better understanding.                                        |

### UNIT 3 [This unit is about Aquaculture Production and Financial management].

|                                                    |                                                                                                                                                                                                                                                                |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit No.</b>                                    | <b>03</b>                                                                                                                                                                                                                                                      |
| <b>Unit Title</b>                                  | <b>Aquaculture Production and Financial management</b>                                                                                                                                                                                                         |
| <b>Description</b>                                 | This unit is about understanding aspects such as the management process itself, aquacultural planning and the principles of financing.                                                                                                                         |
| <b>Scope</b>                                       | This unit covers the following: <ul style="list-style-type: none"> <li>• Aquacultural Project Planning and Appraisal</li> <li>• Project Monitoring and Evaluation</li> <li>• Aquacultural Finance</li> </ul>                                                   |
| <b>Performance Criteria (PC) w.r.t. the Scope</b>  |                                                                                                                                                                                                                                                                |
| <b>Element</b>                                     | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                               |
| <b>Aquacultural project planning and appraisal</b> | To be competent, the individual must be able to:<br>PC1. Develop aquacultural projects<br>PC2. Apply techniques (or investment criteria) for choosing among competing projects<br>PC3. Apply knowledge and skills necessary for optimally allocating resources |
| <b>Project monitoring and evaluation</b>           | To be competent, the individual must be able to:<br>PC4. Apply the key concepts of project monitoring and evaluation in any project setup.<br>PC5. Demonstrate their ability to develop a monitoring and evaluation system                                     |
| <b>Aquacultural finance</b>                        | To be competent, the individual must be able to:<br>PC6. Determine the financial viability of agricultural projects/programmes                                                                                                                                 |

| <b>Knowledge and Understanding (K)</b>                                                      |                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A. Organisational Context (Knowledge of the company/ organisation and its processes)</b> | The individual on the job must demonstrate knowledge and understanding of:<br>OK1. Company regulations and global best practices<br>OK2. Guidelines for liaising with industry players<br>OK3. Code of business conduct<br>OK4. Job responsibilities/duties     |
| <b>B. Technical Knowledge</b>                                                               | The individual on the job must demonstrate knowledge and understanding of:<br>TK1. Aquacultural organisation and administration<br>TK2. Project planning, appraisal, monitoring and evaluation<br>TK3. Aquacultural farm management and finance                 |
| <b>C. Regulatory context (Knowledge of Rules and Regulations)</b>                           | The individual on the job must demonstrate knowledge and understanding of :<br>RK1. Government regulatory requirements in the aquacultural sector                                                                                                               |
| <b>Skills (S)</b>                                                                           |                                                                                                                                                                                                                                                                 |
| <b>A. Core Skills/ Generic Skills</b>                                                       | <b>Writing Skills</b>                                                                                                                                                                                                                                           |
|                                                                                             | The individual on the job must be able to:<br>CS1. Write in English, prepare and submit comprehensive and consolidated reports.                                                                                                                                 |
|                                                                                             | <b>Reading Skills</b>                                                                                                                                                                                                                                           |
|                                                                                             | The individual on the job must be able to:<br>CS2. Read and interpret internal/external documents.<br>CS3. Read and understand manuals, memos, other company documents.<br>CS4. Read from different sources- books, screens in machines and signage.            |
|                                                                                             | <b>Oral Communication (Listening and Speaking skills)</b>                                                                                                                                                                                                       |
| <b>B. Professional Skills</b>                                                               | The individual on the job must be able to:<br>CS5. Express statements or information clearly so that others can hear and understand.<br>CS6. Participate in and understand the main points of simple discussions.<br>CS7. Respond appropriately to any queries. |
|                                                                                             | <b>Decision Making</b>                                                                                                                                                                                                                                          |
|                                                                                             | The individual on the job must be able to:<br>PS1. Follow organization rule-based decision-making process.<br>PS2. Take decision with systematic course of actions and/or response.                                                                             |
|                                                                                             | <b>Plan and Organise</b>                                                                                                                                                                                                                                        |
|                                                                                             | The individual on the job must be able to:<br>PS3. Plan and organise work to meet deadlines.<br>PS4. Work constructively and collaboratively with others.                                                                                                       |
|                                                                                             | <b>Customer Centricity</b>                                                                                                                                                                                                                                      |
|                                                                                             | The individual on the job must be able to:                                                                                                                                                                                                                      |

|  |                                                                                                                                                                                                                                                                          |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | PS5. Follow the code of conduct.<br>PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.                                                                                                                            |
|  | <b>Problem Solving and Decision Making</b>                                                                                                                                                                                                                               |
|  | The individual on the job must be able to:<br>PS7 Recognize problems and search for solutions.<br>PS8. Choose best methods to complete assigned tasks.<br>PS9. Approach relevant authority when required.<br>PS10. Judiciously use common sense in day to day activities |
|  | <b>Analytical Thinking</b>                                                                                                                                                                                                                                               |
|  | The individual on the job must be able to:<br>PS11. Apply domain knowledge, observations and data to select course of action to perform tasks                                                                                                                            |
|  | <b>Critical Thinking</b>                                                                                                                                                                                                                                                 |
|  | The individual on the job must be able to:<br>PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities.<br>PS13. Ask relevant questions for better understanding.                                       |

**UNIT 4** [This unit is about Aquacultural Marketing and Aqua Business].

|                                                   |                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit No.</b>                                   | <b>04</b>                                                                                                                                                                                                                                                                                                                                     |
| <b>Unit Title</b>                                 | <b>Aquaculture Marketing and Aqua Business</b>                                                                                                                                                                                                                                                                                                |
| <b>Description</b>                                | This unit is about understanding all aspects of marketing such as the price system and market types.                                                                                                                                                                                                                                          |
| <b>Scope</b>                                      | This unit covers the following: <ul style="list-style-type: none"> <li>• Aquacultural marketing and pricing</li> <li>• Aqua business management</li> <li>• Aqua business evaluation and planning</li> </ul>                                                                                                                                   |
| <b>Performance Criteria (PC) w.r.t. the Scope</b> |                                                                                                                                                                                                                                                                                                                                               |
| <b>Element</b>                                    | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                              |
| <b>Aquacultural marketing and pricing</b>         | To be competent, the individual must be able to:<br>PC1. Explain concepts for aquacultural commodities<br>PC2. Explain factors that are critical to price variability of aquaculture products                                                                                                                                                 |
| <b>Aqua business management</b>                   | To be competent, the individual must be able to:<br>PC3. Explain the difference between the management role in a small and large aqua business.<br>PC4. Demonstrate the importance of reporting of financial statements in sound financial management<br>PC5. Identify key financial issues faced by the entrepreneurial small agribusinesses |
| <b>Aqua business evaluation and planning</b>      | To be competent, the individual must be able to:<br>PC6. Develop a successful business plan<br>PC7. Develop financial statements and how they can be used to make sound financial decisions for their business<br>PC8. Developing an operation and cash flow budget<br>PC9. Assess the feasibility of a business idea                         |

| <b>Knowledge and Understanding (K)</b>                                                     |                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A. Organisational Context (Knowledge of the company/organisation and its processes)</b> | <p>The individual on the job must demonstrate knowledge and understanding of:</p> <ul style="list-style-type: none"> <li>OK1. Company regulations and global best practices</li> <li>OK2. Guidelines for liaising with industry players</li> <li>OK3. Code of business conduct</li> <li>OK4. Job responsibilities/duties</li> </ul> |
| <b>B. Technical Knowledge</b>                                                              | <p>The individual on the job must demonstrate knowledge and understanding of:</p> <ul style="list-style-type: none"> <li>TK1. Marketing and entrepreneurship skills</li> </ul>                                                                                                                                                      |
| <b>C. Regulatory context (Knowledge of Rules and Regulations)</b>                          | <p>The individual on the job must demonstrate knowledge and understanding of :</p> <ul style="list-style-type: none"> <li>RK1. Government regulatory requirements in the agricultural sector</li> </ul>                                                                                                                             |
| <b>Skills (S)</b>                                                                          |                                                                                                                                                                                                                                                                                                                                     |
| <b>A. Core Skills/ Generic Skills</b>                                                      | <b>Writing Skills</b>                                                                                                                                                                                                                                                                                                               |
|                                                                                            | <p>The individual on the job must be able to:</p> <ul style="list-style-type: none"> <li>CS1. Write in English, prepare and submit comprehensive and consolidated reports.</li> </ul>                                                                                                                                               |
|                                                                                            | <b>Reading Skills</b>                                                                                                                                                                                                                                                                                                               |
|                                                                                            | <p>The individual on the job must be able to:</p> <ul style="list-style-type: none"> <li>CS2. Read and interpret internal/external documents.</li> <li>CS3. Read and understand manuals, memos, other company documents.</li> <li>CS4. Read from different sources- books, screens in machines and signage.</li> </ul>              |
|                                                                                            | <b>Oral Communication (Listening and Speaking skills)</b>                                                                                                                                                                                                                                                                           |
|                                                                                            | <p>The individual on the job must be able to :</p> <ul style="list-style-type: none"> <li>CS5. Express statements or information clearly so that others can hear and understand.</li> <li>CS6. Participate in and understand the main points of simple discussions.</li> <li>CS7. Respond appropriately to any queries.</li> </ul>  |
| <b>B. Professional Skills</b>                                                              | <b>Decision Making</b>                                                                                                                                                                                                                                                                                                              |
|                                                                                            | <p>The individual on the job must be able to:</p> <ul style="list-style-type: none"> <li>PS1. Follow organization rule-based decision-making process.</li> <li>PS2. Take decision with systematic course of actions and/or response.</li> </ul>                                                                                     |
|                                                                                            | <b>Plan and Organise</b>                                                                                                                                                                                                                                                                                                            |
|                                                                                            | <p>The individual on the job must be able to:</p> <ul style="list-style-type: none"> <li>PS3. Plan and organise work to meet deadlines.</li> <li>PS4. Work constructively and collaboratively with others.</li> </ul>                                                                                                               |
|                                                                                            | <b>Customer Centricity</b>                                                                                                                                                                                                                                                                                                          |
|                                                                                            | <p>The individual on the job must be able to:</p> <ul style="list-style-type: none"> <li>PS5. Follow the code of conduct.</li> <li>PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.</li> </ul>                                                                             |
|                                                                                            | <b>Problem Solving and Decision Making</b>                                                                                                                                                                                                                                                                                          |

|  |                                                                                                                                                                                                                                                                           |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | The individual on the job must be able to:<br>PS7. Recognize problems and search for solutions.<br>PS8. Choose best methods to complete assigned tasks.<br>PS9. Approach relevant authority when required.<br>PS10. Judiciously use common sense in day to day activities |
|  | <b>Analytical Thinking</b>                                                                                                                                                                                                                                                |
|  | The individual on the job must be able to:<br>PS11. Apply domain knowledge, observations and data to select course of action to perform tasks                                                                                                                             |
|  | <b>Critical Thinking</b>                                                                                                                                                                                                                                                  |
|  | The individual on the job must be able to:<br>PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities.                                                                                                  |
|  | PS13. Ask relevant questions for better understanding.                                                                                                                                                                                                                    |

**UNIT 5** [This unit is about Aquacultural Policy Analysis and Design].

|                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit No.</b>                                                                             | <b>05</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Unit Title</b>                                                                           | <b>Aquacultural Policy Analysis and Design</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Description</b>                                                                          | This unit is about understanding the interaction between aquaculture and trade policy, production policy, price and income policy and the government functions.                                                                                                                                                                                                                                                                                                         |
| <b>Scope</b>                                                                                | This unit covers the following: <ul style="list-style-type: none"> <li>• Aquacultural policy analysis</li> </ul>                                                                                                                                                                                                                                                                                                                                                        |
| <b>Performance Criteria (PC) w.r.t. the Scope</b>                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Element</b>                                                                              | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Aquacultural policy analysis</b>                                                         | To be competent, the individual must be able to:<br>PC1. Identify the goals and objectives of aquacultural policy<br>PC2. Recognize the alternative policy instruments for government intervention<br>PC3. Apply various tools and approaches of aquacultural policy analysis<br>PC4. Assess the efficiency of government institutions and the private sector in aqua cultural research<br>PC5. Outline the relevance and applications of aquacultural policy analysis. |
| <b>Knowledge and Understanding (K)</b>                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>A. Organisational Context (Knowledge of the company/ organisation and its processes)</b> | The individual on the job must demonstrate knowledge and understanding of:<br>OK1. Company regulations and global best practices<br>OK2. Guidelines for liaising with industry players<br>OK3. Code of business conduct<br>OK4. Job responsibilities/duties                                                                                                                                                                                                             |

|                                                                   |                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>B. Technical Knowledge</b>                                     | The individual on the job must demonstrate knowledge and understanding of:<br>TK1. Policy analysis<br>TK2. Policy designs                                                                                                                                                 |
| <b>C. Regulatory context (Knowledge of Rules and Regulations)</b> | The individual on the job must demonstrate knowledge and understanding of :<br>RK1. Government regulatory requirements in the aquacultural sector.                                                                                                                        |
| <b>Skills (S)</b>                                                 |                                                                                                                                                                                                                                                                           |
| <b>A. Core Skills/ Generic Skills</b>                             | <b>Writing Skills</b>                                                                                                                                                                                                                                                     |
|                                                                   | The individual on the job must be able to:<br>CS1. Write in English, prepare and submit comprehensive and consolidated reports.                                                                                                                                           |
|                                                                   | <b>Reading Skills</b>                                                                                                                                                                                                                                                     |
|                                                                   | The individual on the job must be able to:<br>CS2. Read and interpret internal/external documents.<br>CS3. Read and understand manuals, memos, other company documents.<br>CS4. Read from different sources- books, screens in machines and signage.                      |
|                                                                   | <b>Oral Communication (Listening and Speaking skills)</b>                                                                                                                                                                                                                 |
|                                                                   | The individual on the job must be able to :<br>CS5. Express statements or information clearly so that others can hear and understand.<br>CS6. Participate in and understand the main points of simple discussions.<br>CS7. Respond appropriately to any queries.          |
| <b>B. Professional Skills</b>                                     | <b>Decision Making</b>                                                                                                                                                                                                                                                    |
|                                                                   | The individual on the job must be able to:<br>PS1. Follow organization rule-based decision-making process.<br>PS2. Take decision with systematic course of actions and/or response.                                                                                       |
|                                                                   | <b>Plan and Organise</b>                                                                                                                                                                                                                                                  |
|                                                                   | The individual on the job must be able to:<br>PS3. Plan and organise work to meet deadlines.<br>PS4. Work constructively and collaboratively with others.                                                                                                                 |
|                                                                   | <b>Customer Centricity</b>                                                                                                                                                                                                                                                |
|                                                                   | The individual on the job must be able to:<br>PS5. Follow the code of conduct.<br>PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.                                                                               |
|                                                                   | <b>Problem Solving and Decision Making</b>                                                                                                                                                                                                                                |
|                                                                   | The individual on the job must be able to:<br>PS7. Recognize problems and search for solutions.<br>PS8. Choose best methods to complete assigned tasks.<br>PS9. Approach relevant authority when required.<br>PS10. Judiciously use common sense in day to day activities |
|                                                                   | <b>Analytical Thinking</b>                                                                                                                                                                                                                                                |

|  |                                                                                                                                                                                                                                    |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | The individual on the job must be able to:<br>PS11. Apply domain knowledge, observations and data to select course of action to perform tasks                                                                                      |
|  | <b>Critical Thinking</b>                                                                                                                                                                                                           |
|  | The individual on the job must be able to:<br>PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities.<br>PS13. Ask relevant questions for better understanding. |

**UNIT 6** [This unit is about Aquacultural Development and Environmental Economics].

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit No.</b>                                   | <b>06</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Unit Title</b>                                 | <b>Aquacultural Development and Environmental Economics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Description</b>                                | This unit is about the environmental sustainability of aquaculture activities that lead to the degradation or removal of ecologically valuable habitats.<br>It is also concerned with introductions and transmission of pathogens and genes between cultured and wild stocks.<br>Further, it addresses economic, social and cultural sustainability concerns including dramatic production losses caused by epidemic outbreaks of diseases and cultural impacts of conversion from agricultural multi-crops to pond culture.              |
| <b>Scope</b>                                      | This unit covers the following: <ul style="list-style-type: none"> <li>• Development Economics</li> <li>• Environmental economics</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Performance Criteria (PC) w.r.t. the Scope</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Element</b>                                    | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Development economics</b>                      | To be competent, the individual must be able to:<br>PC1. Outline the nature, scope and methodology of development economics.<br>PC2. Depict the basis for applied and advanced study of development economics.                                                                                                                                                                                                                                                                                                                            |
| <b>Environmental economics</b>                    | To be competent, the individual must be able to:<br>PC3. Analyse environmental problems such as air and water pollution, deforestation, soil degradation, and sedimentation.<br>PC4. Identify specific environmental problems in the aquaculture development.<br>PC5. Employ the approach of environmental economics to the solution of environmental problems.<br>PC6. Identify diseases and pathogens prevalent to Aquaculture<br>PC7. Identify Economic, social and cultural sustainability concerns and describe mitigation measures. |

| <b>Knowledge and Understanding (K)</b>                                                         |                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A. Organisational Context<br/>(Knowledge of the company/organisation and its processes)</b> | The individual on the job must demonstrate knowledge and understanding of:<br>OK1. Company regulations and global best practices<br>OK2. Guidelines for liaising with industry players<br>OK3. Code of business conduct<br>OK4. Job responsibilities/duties      |
| <b>B. Technical Knowledge</b>                                                                  | The individual on the job must demonstrate knowledge and understanding of:<br>TK1. Developmental models<br>TK2. Economic/environmental models                                                                                                                    |
| <b>C. Regulatory context<br/>(Knowledge of Rules and Regulations)</b>                          | The individual on the job must demonstrate knowledge and understanding of :<br>RK1. Government regulatory requirements in the agricultural sector                                                                                                                |
| <b>Skills (S)</b>                                                                              |                                                                                                                                                                                                                                                                  |
| <b>A. Core Skills/<br/>Generic Skills</b>                                                      | <b>Writing Skills</b>                                                                                                                                                                                                                                            |
|                                                                                                | The individual on the job must be able to:<br>CS1. Write in English, prepare and submit comprehensive and consolidated reports.                                                                                                                                  |
|                                                                                                | <b>Reading Skills</b>                                                                                                                                                                                                                                            |
|                                                                                                | The individual on the job must be able to:<br>CS2. Read and interpret internal/external documents.<br>CS3. Read and understand manuals, memos, other company documents.<br>CS4. Read from different sources- books, screens in machines and signage.             |
|                                                                                                | <b>Oral Communication (Listening and Speaking skills)</b>                                                                                                                                                                                                        |
| <b>B. Professional Skills</b>                                                                  | The individual on the job must be able to :<br>CS5. Express statements or information clearly so that others can hear and understand.<br>CS6. Participate in and understand the main points of simple discussions.<br>CS7. Respond appropriately to any queries. |
|                                                                                                | <b>Decision Making</b>                                                                                                                                                                                                                                           |
|                                                                                                | The individual on the job must be able to:<br>PS1. Follow organization rule-based decision-making process.<br>PS2. Take decision with systematic course of actions and/or response.                                                                              |
|                                                                                                | <b>Plan and Organise</b>                                                                                                                                                                                                                                         |
|                                                                                                | The individual on the job must be able to:<br>PS3. Plan and organise work to meet deadlines.<br>PS4. Work constructively and collaboratively with others.                                                                                                        |
| <b>B. Professional Skills</b>                                                                  | <b>Customer Centricity</b>                                                                                                                                                                                                                                       |
|                                                                                                | The individual on the job must be able to:<br>PS5. Follow the code of conduct.                                                                                                                                                                                   |

|  |                                                                                                                                                                                                                                                                          |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.                                                                                                                                                                |
|  | <b>Problem Solving and Decision Making</b>                                                                                                                                                                                                                               |
|  | The individual on the job must be able to:<br>PS7 Recognize problems and search for solutions.<br>PS8. Choose best methods to complete assigned tasks.<br>PS9. Approach relevant authority when required.<br>PS10. Judiciously use common sense in day to day activities |
|  | <b>Analytical Thinking</b>                                                                                                                                                                                                                                               |
|  | The individual on the job must be able to:<br>PS11. Apply domain knowledge, observations and data to select course of action to perform tasks                                                                                                                            |
|  | <b>Critical Thinking</b>                                                                                                                                                                                                                                                 |
|  | The individual on the job must be able to:<br>PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities.<br>PS13. Ask relevant questions for better understanding.                                       |

**UNIT 7** [This unit is about Aquacultural Economics Operational Research].

|                                                       |                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit No.</b>                                       | <b>07</b>                                                                                                                                                                                                                                                                                                                               |
| <b>Unit Title</b>                                     | <b>Aquacultural Economics and Operational Research</b>                                                                                                                                                                                                                                                                                  |
| <b>Description</b>                                    | This unit is about understanding various tools and the application of economic simulation and optimisation techniques on aquacultural problems.<br>In operations research, problems are broken down into basic components and then solved in defined steps by mathematical analysis, (Implementing the solution to the actual problem). |
| <b>Scope</b>                                          | This unit covers the following: <ul style="list-style-type: none"> <li>• Introductory statistics in Aquaculture</li> <li>• Quantitative Methods in Aquacultural Economics</li> <li>• Communication &amp; Research Methods</li> <li>• Research Projects</li> </ul>                                                                       |
| <b>Performance Criteria (PC) w.r.t. the Scope</b>     |                                                                                                                                                                                                                                                                                                                                         |
| <b>Element</b>                                        | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                        |
| <b>Introductory statistics for aquaculture</b>        | To be competent, the individual must be able to:<br>PC1. Use statistics in order to gauge the probability of a particular outcome.                                                                                                                                                                                                      |
| <b>Quantitative Methods in Aquacultural Economics</b> | To be competent, the individual must be able to:<br>PC2. Implement appropriate techniques in resource management                                                                                                                                                                                                                        |
| <b>Communication and Research methods</b>             | To be competent, the individual must be able to:<br>PC3. Apply communication concepts and skills<br>PC4. Apply key concepts of professional and scientific writing                                                                                                                                                                      |

|                                                                                            |                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Research Projects</b>                                                                   | To be competent, the individual must be able to:<br>PC5. Develop and present a research proposal.<br>PC6. Implement and write research projects.                                                                                                                 |
| <b>Knowledge and Understanding (K)</b>                                                     |                                                                                                                                                                                                                                                                  |
| <b>A. Organisational Context (Knowledge of the company/organisation and its processes)</b> | The individual on the job must demonstrate knowledge and understanding of:<br>OK1. Company regulations and global best practices<br>OK2. Guidelines for liaising with industry players<br>OK3. Code of business conduct<br>OK4. Job responsibilities/duties      |
| <b>B. Technical Knowledge</b>                                                              | The individual on the job must demonstrate knowledge and understanding of:<br>TK1. Various analytical models                                                                                                                                                     |
| <b>C. Regulatory context (Knowledge of Rules and Regulations)</b>                          | The individual on the job must demonstrate knowledge and understanding of :<br>RK1. Government regulatory requirements in the aquaculture sector                                                                                                                 |
| <b>Skills (S)</b>                                                                          |                                                                                                                                                                                                                                                                  |
| <b>A. Core Skills/ Generic Skills</b>                                                      | <b>Writing Skills</b>                                                                                                                                                                                                                                            |
|                                                                                            | The individual on the job must be able to:<br>CS1. Write in English, prepare and submit comprehensive and consolidated reports.                                                                                                                                  |
|                                                                                            | <b>Reading Skills</b>                                                                                                                                                                                                                                            |
|                                                                                            | The individual on the job must be able to:<br>CS2. Read and interpret internal/external documents.<br>CS3. Read and understand manuals, memos, and other company documents.<br>CS4. Read from different sources- books, screens in machines and signage.         |
|                                                                                            | <b>Oral Communication (Listening and Speaking skills)</b>                                                                                                                                                                                                        |
| <b>B. Professional Skills</b>                                                              | The individual on the job must be able to :<br>CS5. Express statements or information clearly so that others can hear and understand.<br>CS6. Participate in and understand the main points of simple discussions.<br>CS7. Respond appropriately to any queries. |
|                                                                                            | <b>Decision Making</b>                                                                                                                                                                                                                                           |
|                                                                                            | The individual on the job must be able to:<br>PS1. Follow organization rule-based decision-making process.<br>PS2. Take decisions with a systematic course of action and/or response.                                                                            |
|                                                                                            | <b>Plan and Organise</b>                                                                                                                                                                                                                                         |
|                                                                                            | The individual on the job must be able to:<br>PS3. Plan and organise work to meet deadlines.                                                                                                                                                                     |

|  |                                                                                                                            |
|--|----------------------------------------------------------------------------------------------------------------------------|
|  | PS4. Work constructively and collaboratively with others.                                                                  |
|  | <b>Customer Centricity</b>                                                                                                 |
|  | The individual on the job must be able to:                                                                                 |
|  | PS5. Follow the code of conduct.                                                                                           |
|  | PS6. Manage relationships with customers with the intent of satisfying its requirements for service delivery.              |
|  | <b>Problem Solving and Decision Making</b>                                                                                 |
|  | The individual on the job must be able to:                                                                                 |
|  | PS7. Recognize problems and search for solutions.                                                                          |
|  | PS8. Choose the best methods to complete assigned tasks.                                                                   |
|  | PS9. Approach relevant authority when required.                                                                            |
|  | PS10. Judiciously use common sense in day to day activities.                                                               |
|  | <b>Analytical Thinking</b>                                                                                                 |
|  | The individual on the job must be able to:                                                                                 |
|  | PS11. Apply domain knowledge, observations and data to select course of action to perform tasks                            |
|  | <b>Critical Thinking</b>                                                                                                   |
|  | The individual on the job must be able to:                                                                                 |
|  | PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities. |
|  | PS13. Ask relevant questions for better understanding.                                                                     |

## **5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS**

These include, but not limited to; Computers, Printer and relevant Software programs, Personal protective equipment (PPE) and various laboratory equipment and consumables.

## **6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER**

Dilemmas associated with the job of an Aquacultural Economist include dealing with data from aquaculture farms that is not well recorded because of poor records and non-existent of data in some cases. Data weaknesses may have implications on the development of aquacultural policies.

### **6.1 Alternative Choices (Solutions) to Dilemmas and Complexities**

Solutions to dilemmas include, intensifying trainings to aquacultural farmers in proper recording keeping and management of their farms.

## **7. WORKING CONDITIONS/ENVIRONMENT**

Working conditions include working in the field under varied weather conditions, in the office for long hours, working areas that are sometimes noisy.

## **8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE**

### **8.1 Internal/Within the Organization**

Parties involved/interacting with the job holder who are internal to the organization include supervisors, subordinates, and other employees.

### **8.2 External/Outside the Organization**

Parties involved/interacting with the job holder who are external include; non-governmental organisations and donor agencies on matters of programme planning and development, customers/clients, government regulators, trainers, suppliers of equipment/tools/consumables, academia etc.

## **9. PHYSICAL DEMANDS ON THE BODY**

- Minimum physical effort for both office and field work.
- Be able to walk and stand for some periods of time;
- Substantial mental effort associated with preparation of technical reports;
- Be able to use fingers, hands and feet with ease to complete the assigned task (dexterity);
- Etc.

## **ANNEX A**

### **Criteria for Assessments based on this NOS**

#### **A.1 Guidelines for Assessment**

**A.1.1** Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programmes developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programmes developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

**A.1.2** Awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part and evaluations for skill practical part for their respective candidates.

## ANNEX B

### NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

|                   |                        |                            |                           |
|-------------------|------------------------|----------------------------|---------------------------|
| <b>NOS Code</b>   | NOS.AE.01              |                            |                           |
| <b>ZQF Level</b>  | 7                      | <b>Version Number</b>      | 01                        |
| <b>Sector</b>     | Agriculture            | <b>Date of Approval</b>    | 19 <sup>th</sup> May 2022 |
| <b>Sub Sector</b> |                        | <b>Date of Last Review</b> | N/A                       |
| <b>Occupation</b> | Aquacultural Economist | <b>Date of Next Review</b> | May 2027                  |

## **REGISTERED OFFICE**

Zambia Qualifications Authority  
Ground Floor, Finsbury Park  
P.O Box 51103  
Lusaka, Zambia

**Tel:** +260 211 843050/ 843053

**Mobile:** +260 963 922 730/ 0956 037 185 / 0972 559 301

**Email:** [info@zaqa.gov.zm](mailto:info@zaqa.gov.zm)

**Website:** [www.zaqa.gov.zm](http://www.zaqa.gov.zm)